

Solution 5

q.1

searching on blastp with nr (clustered)

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blastn

blastp

blastx

tblastn

tblastx

Standard Protein BLAST

BLASTP programs search protein databases using a protein query. [more...](#)

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Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#)

>1336093|Genbank|Outer membrane integral membrane protein|HrcC
MVEKRELRCRLGALLMLCATLPAGAQTADWKEQSYAYSADRTPLSTVLQD
FADGHSVD
LHLGNVEDTEVTAKIRAENASAFDLRLALEHHFQWFVYNNNTLYVSPQDEQSS

Query subrange [?](#)

From

To

Or, upload file

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Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database

ClusteredNR (clustered nr) [?](#)

Organism

Optional

[Add organism](#)

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown. [?](#)

Program Selection

Algorithm

☒ blastp (protein-protein BLAST)

☐ PSI-BLAST (Position-Specific Iterated BLAST)

Choose a BLAST algorithm [?](#)

Search database ClusteredNR using Blastp (protein-protein BLAST)

☐ Show results in a new window

[+ Algorithm parameters](#)

BLAST® » blastp suite » results for RID-9ZXP1P4M014

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Job Title 1336093|Genbank|Outer membrane integral
 RID [9ZXP1P4M014](#) Search expires on 09-09 19:51 pm [Download All](#)
 Program BLASTP [Citation](#)
 Database [ClusteredNR](#) [See details](#)
 Query ID Icd|Query_7180130
 Description 1336093|Genbank|Outer membrane integral membrane pr ...
 Molecule type amino acid
 Query Length 676
 Other reports [Distance tree of results](#) [Multiple alignment](#) [MSA viewer](#)

Filter Results

Organism only top 20 will appear
 Type common name, binomial, taxid or group name
[+ Add organism](#)

Percent Identity to E value to Query Coverage to
[Filter](#) [Reset](#)

Clusters Graphic Summary Alignments Taxonomy

Clusters producing significant alignments Download Select columns Show 100

☒ select all 100 clusters selected [GenPept](#) [Graphics](#) [Distance tree of results](#) [Multiple alignment](#) [MSA Viewer](#)

Cluster Composition	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
12 member(s), 6 organism(s)	enterobacteria	type III secretion system outer membrane ring subunit SctC [Erwinia a...	1371	1371	100%	0.0	98.97%	677	WP_004168436...
3 member(s), 2 organism(s)	enterobacteria	type III secretion system outer membrane ring subunit SctC [Erwinia tr...	1183	1183	100%	0.0	83.70%	674	WP_016191026...
18 member(s), 10 organism(s)	enterobacteria	type III secretion system outer membrane ring subunit SctC [Erwinia ...	1163	1163	100%	0.0	84.24%	679	WP_034936517...
15 member(s), 4 organism(s)	enterobacteria	type III secretion system outer membrane ring subunit SctC [Pantoea ...	1148	1148	100%	0.0	81.66%	677	WP_039339364...
4 member(s), 3 organism(s)	enterobacteria	type III secretion system outer membrane ring subunit SctC [Pantoea ...	1137	1137	100%	0.0	81.54%	677	WP_064703208...
2 member(s), 1 organism(s)	enterobacteria	type III secretion system outer membrane ring subunit SctC [Enteroba...	1108	1108	100%	0.0	82.69%	674	WP_491670091...

no. of occurrences or clusters = 100 here
 and max score of 1371 with 100 percent query coverage
 in swissprot there are lots of

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Job Title 1336093|Genbank|Outer membrane integral
 RID [9ZY0RVV014](#) Search expires on 09-09 20:12 pm [Download All](#)
 Program BLASTP [Citation](#)
 Database [swissprot](#) [See details](#)
 Query ID Icd|Query_12271979
 Description 1336093|Genbank|Outer membrane integral membrane pr ...
 Molecule type amino acid
 Query Length 676
 Other reports [Distance tree of results](#) [Multiple alignment](#) [MSA viewer](#)

Filter Results

Organism only top 20 will appear ☐ exclude
 Type common name, binomial, taxid or group name
[+ Add organism](#)

Percent Identity to E value to Query Coverage to
[Filter](#) [Reset](#)

Descriptions Graphic Summary Alignments Taxonomy

Sequences producing significant alignments Download Select columns Show 100

☒ select all 32 sequences selected [GenPept](#) [Graphics](#) [Distance tree of results](#) [Multiple alignment](#) [MSA Viewer](#)

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
RecName: Full=Type 3 secretion system secretin; Short=T3SS secretin; AltName: Full=Hypersensitivity re...	Pseudomonas...	566	566	99%	0.0	44.16%	701	Q01723.2
RecName: Full=Type 3 secretion system secretin; Short=T3SS secretin; AltName: Full=YscC secretin; Fla...	Yersinia entero...	251	251	72%	3e-73	31.25%	607	Q01244.1
RecName: Full=Type 3 secretion system secretin; Short=T3SS secretin; AltName: Full=Type III secretion p...	Yersinia pestis	246	246	76%	1e-71	30.52%	607	Q56974.1
RecName: Full=Type 3 secretion system secretin; Short=T3SS secretin; AltName: Full=Hypersensitivity re...	Ralstonia pseu...	211	211	75%	5e-59	28.87%	568	Q52498.1
RecName: Full=Type 3 secretion system secretin; Short=T3SS secretin; AltName: Full=Outer membrane p...	Shigella flexner	175	175	70%	6e-46	27.63%	566	Q04641.1
RecName: Full=Type 3 secretion system secretin; Short=T3SS secretin; AltName: Full=Outer membrane p...	Shigella sonnei	175	175	70%	6e-46	27.63%	566	Q55293.1
RecName: Full=SPI-2 type 3 secretion system secretin; Short=T3SS-2 secretin; AltName: Full=Outer mem...	Salmonella ent...	158	158	73%	2e-40	26.13%	497	P02WR9.1
RecName: Full=Type 3 secretion system secretin; Short=T3SS secretin; AltName: Full=Hypersensitivity re...	Xanthomonas...	132	132	37%	5e-31	31.25%	607	P80151.1

there are only 32 sequences with max score of 566 only, and query coverage is 99 percent highest

Q.2

the algorithm parameters

The screenshot shows the 'General Parameters' section of the NCBI BLAST interface. It includes a 'Restore default search parameters' button in the top right. The 'General Parameters' section contains: 'Max target sequences' set to 100, 'Short queries' checked with 'Automatically adjust parameters for short input sequences', 'Expect threshold' set to 0.05, 'Word size' set to 5, and 'Max matches in a query range' set to 0. The 'Scoring Parameters' section includes: 'Matrix' set to BLOSUM62, 'Gap Costs' set to 'Existence: 11 Extension: 1', and 'Compositional adjustments' set to 'Conditional compositional score matrix adjustment'. The 'Filters and Masking' section has three unchecked options: 'Low complexity regions', 'Mask for lookup table only', and 'Mask lower case letters'. At the bottom, there is a 'BLAST' button and a search bar containing 'Search database ClusteredNR using Blastp (protein-protein BLAST)'. A 'Show results in a new window' checkbox is also present. A 'feedback' button is visible on the right side.

E threshold of 0.05, word size of 5
there are total 5 main parameters and more are also given

Q.3

given the query sequence of hrcc and the accession number for the other

The screenshot shows the NCBI BLAST interface with the 'blastp' algorithm selected. The 'Enter Query Sequence' section contains a text box with the sequence: >1336093(Genbank)Outer membrane integral membrane proteinHrcC
MVEKRELRCRLGALLMLCATLPAGATPADWKEQSYAYSADRTPLSTVLQD
FIADGHSDV
LHLGNVEDTEVTAKIRAENASAFDLRLALEHHFQWVYNNITLVSPQDEQSS. Below this is a 'Job Title' field and a checkbox for 'Align two or more sequences'. The 'Enter Subject Sequence' section contains a text box with the accession number: AA881929.1. Below this is a 'Job Title' field and a checkbox for 'Align two or more sequences'. The 'Program Selection' section shows 'blastp (protein-protein BLAST)' selected. At the bottom, there is a 'BLAST' button and a search bar containing 'Search using Blastp (protein-protein BLAST)'. A 'Show results in a new window' checkbox is also present. A '+ Algorithm parameters' button is visible at the bottom left.

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blastn **blastp** blastx tblastn tblastx Align Sequences Protein BLAST

BLAST programs search protein subjects using a protein query, more...

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) Clear Query subrange [?](#)

PDIAVGNPK
VKAHGKKVLAFTSDGLAHLDNLKGTFTLSELHCKLHVDPENFLGGVIV
CYLAHFG
KEFTPRQAAVQKVVAGVANALAHKYH

From To

Or, upload file No file chosen [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☒ Align two or more sequences [?](#)

Enter Subject Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) Clear Subject subrange [?](#)

TAIGNPM
VRAHGKVLTSFGDAVKNLNKKITFSQLSELHCKLHVDPENFLGGVIV
LAARFS
KDFTPQAAVQKVVAGVANALAHKYH

From To

Or, upload file No file chosen [?](#)

Program Selection

Algorithm ☒ blastp (protein-protein BLAST)
[Choose a BLAST algorithm](#) [?](#)

Search protein sequence using Blastp (protein-protein BLAST)
☐ Show results in a new window

Note: Parameter values that differ from the default are highlighted in yellow and marked with * sign

+ Algorithm parameters

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Job Title sp|P68871|HBB_HUMAN Hemoglobin subunit

RID [A00058PR114](#) Search expires on 09-09 20:31 pm [Download All](#) [?](#)

Program Blast 2 sequences [Citation](#) [?](#)

Query ID Icl|Query_8380397 (amino acid)

Query Descr sp|P68871|HBB_HUMAN Hemoglobin subunit beta OS=Hc ...

Query Length 147

Subject ID Icl|Query_8380399 (amino acid)

Subject Descr sp|P02112|HBB_CHICK Hemoglobin subunit beta OS=Gall ...

Subject Length 147

Other reports [Multiple alignment](#) [MSA viewer](#) [?](#)

Clusters Graphic Summary Alignments Dot Plot

Clusters producing significant alignments Download Select columns Show 100 [?](#)

☒ select all 1 clusters selected [Graphics](#) [Multiple alignment](#) [MSA Viewer](#)

	Cluster Representative	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	sp P02112 HBB_CHICK Hemoglobin subunit beta OS=Gallus gallus OX=9031 GN=HBB PE=1 S...	221	221	100%	1e-80	69.39%	147	Query_8380399

- **Percent Identity: 69.39%** (102 / 147 residues identical)
- **Query Coverage: 100%** (full-length alignment with no truncation)
- **Max / Total Score: 221 bits**
- **E-value: 1e-80** (extremely significant, confirming definitive homology rather than random chance)
- **Gaps: 0%** (0 / 147, both proteins align end-to-end without insertions or deletions)
- **Positives (Similarity): ~84%** (accounting for conservative amino acid substitutions with positive BLOSUM62 scores)

Despite the divergence of avian and mammalian lineages approximately 310 million years ago, the sequences share nearly 70%

exact sequence identity and 84% physicochemical similarity.

Q.5

we can use a sliding window of $k=5$ to do this

basically the question is asking to find identical 5 amino acid fragments between 2 protein sequences

for 147 aa we will have $147 - 5 + 1 = 143$ pentapeptides or windows to check for

so simply store a hash map and update the frequency as we move the sliding window for each pentapeptide

code also given in github [link](#)

```
assignment_5 > q5.py ~
1 s1 = "MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAMGNPKVKAHGKKVLGAFSDGLAHDLNLTGTATLSEHCDKLVHDPENFRLLGNVLVCVLAHHFGKEFTPPVQAAAYQKVAVGANALAHKYH"
2 s2 = "MVHMTAEKQLITGLWGKVNVAECGAEALARLLIVYPWTQRFFASFGNLSPTAILGNPMVRAHGKKVLTSFGDAVKNLNDIKNTFAQLSEHCDKLVHDPENFRLLGDILIVLAHFHFKDFTPECQAAMQKLVVRVVAHALAHKYH"
3
4 d1 = {}
5 d2 = {}
6
7 for i in range(len(s1) - 4):
8     p = s1[i:i+5]
9     if p in d1:
10         d1[p] = d1[p] + 1
11     else:
12         d1[p] = 1
13
14 for i in range(len(s2) - 4):
15     p = s2[i:i+5]
16     if p in d2:
17         d2[p] = d2[p] + 1
18     else:
19         d2[p] = 1
20
21 ans = []
22 for k in d1:
23     if k in d2:
24         ans.append(k)
25
26 print("Matching pentapeptides:", len(ans))
27 print("peptide seq1_count seq2_count")
28
29 for x in ans:
30     print(x, "\t", d1[x], "\t", d2[x])
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Kernel: 6.8.0-138-generic
Uptime: 7 days, 3 hours, 11 mins
Shell: zsh 5.8.1

`> /usr/bin/python3 /home/specapoorv/bioinfo/assignment_5/q5.py`
Matching pentapeptides: 38
peptide seq1_count seq2_count
LWGKV 1 1
WGKVN 1 1
GKVVN 1 1
VYPWT 1 1
YPWTQ 1 1
PWITQ 1 1
WTQRF 1 1

Ln 24, Col 22 Spaces: 4 UTF-8 LF {} Python Python 3.10.12.final.0

q.6

[code link](#)

You take the two aligned strings (including - if there are any gaps) and compare them position by position with a single loop:

- **Identity:** Both letters are identical (`a1 == a2` and not a dash).

- **Similarity (Positives):** The letters are identical OR have a positive substitution score (chemically similar, like **K** and **R**, or **I** and **V**).
- **Gaps:** Either sequence has a **-** at that position.
- **Query Coverage:** How many non-gap positions from query sequence made it into the alignment, divided by the full query sequence length

so we can write code for this like this

we will define a group of similar AAs first

```

gnement_5 > q6.py > ...
s1 = "MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAMGNPKVKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLCVLAHFGKEFTPPVQAAQYQKVAGVANALAHKYH"
s2 = "MVHWTAEKOLITGLWGVNVAECGAELARLLIVYPWTQRFASFGNLSPTAILGNPMVRAHGKKVLTSGDAVKNLDNIKNTFAQLSELHCDKLHVDPENFRLLGDILIIIVLAHFSKDFTEPCQAQWOKLVRVVAHALAHKYH"

similar_groups = [
    ["K", "R", "H"],
    ["D", "E"],
    ["I", "L", "V", "M"],
    ["F", "Y", "W"],
    ["S", "T"],
    ["A", "G"]
]

matches = 0
similars = 0
gaps = 0
total_len = len(s1)

for i in range(total_len):
    c1 = s1[i]
    c2 = s2[i]

    if c1 == "-" or c2 == "-":
        gaps = gaps + 1
    elif c1 == c2:
        matches = matches + 1
        similars = similars + 1
    else:
        for g in similar_groups:
            if c1 in g and c2 in g:
                similars = similars + 1
                break

orig_query_len = 147
query_aligned_bases = 0
for ch in s1:
    if ch != "-":
        query_aligned_bases = query_aligned_bases + 1

ident_pct = (matches / total_len) * 100
sim_pct = (similars / total_len) * 100
gap_pct = (gaps / total_len) * 100
cov_pct = (query_aligned_bases / orig_query_len) * 100

> /usr/bin/python3 /home/specapoorv/bioinfo/assignment_5/q6.py
Length: 147
Matches: 102
Similars: 120
Gaps: 0
Identity %: 69.38775510204081
Similarity %: 81.63265306122449
Gap %: 0.0
Query Coverage %: 100.0
~/bioinfo main ?1 >

```

the answer matches the q4's answer, slight difference in similarity could be because of how similarity is defined and we just used a simple group, they might be using blosum matrix etc

Q.7

UniProtKB 3,486 results

Entry	Entry Name	Protein Names	Gene Names	Organism	Length
☒ Q74700	TIM9_YEAST	Mitochondrial import inner membrane translocase subunit TIM9	TIM9, YEL020W-A, YEL020BW	Saccharomyces cerevisiae (strain ATCC 204508 / S288c) (Baker's yeast)	87 AA
☒ Q9Y5J7	TIM9_HUMAN	Mitochondrial import inner membrane translocase subunit Tim9	TIMM9, TIM9, TIM9A, TIMM9A	Homo sapiens (Human)	89 AA
☒ Q9WV98	TIM9_MOUSE	Mitochondrial import inner membrane translocase subunit Tim9	Timm9, Tim9, Tim9a, Timm9a	Mus musculus (Mouse)	89 AA
☒ Q9WV97	TIM9_RAT	Mitochondrial import inner membrane translocase subunit Tim9	Timm9, Tim9, Tim9a, Timm9a	Rattus norvegicus (Rat)	89 AA
☒ Q8J1Z1	TIM9_NEUCR	Mitochondrial import inner membrane translocase subunit tim9	tim9, NCU00198	Neurospora crassa (strain ATCC 24698 / 74-OR23-1A / CBS 708.71 / DSM 1257 / FGSC 987)	88 AA
☒ Q2KIV2	TIM9_BOVIN	Mitochondrial import inner membrane translocase subunit Tim9	TIMM9	Bos taurus (Bovine)	89 AA
☒ Q9XGX9	TIM9_ARATH	Mitochondrial import inner membrane translocase subunit TIM9[...]	TIM9, EMB2474, At3g46560, F12A12.80	Arabidopsis thaliana (Mouse-ear cress)	93 AA
☒ Q9VYD7	TIM9_DROME	Mitochondrial import inner	Tim9a, Tim9, CG1660	Drosophila melanogaster (Fruit fly)	95 AA

selected 20 protein of different organisms

fasta file link

Clustal Omega

pasted the sequence of these into clustal omega

Clustal Omega

Multiple Sequence Alignment (MSA)

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Welcome to the Job Dispatcher website! If you need assistance or have feedback, please [contact](#) us.

Clustal Omega is a new multiple sequence alignment program that uses seeded guide trees and HMM profile-profile techniques to generate alignments between three or more sequences. For the alignment of two sequences please instead use our pairwise sequence alignment tools. This tool can align up to 4000 sequences or a maximum file size of 4 MB.

Input sequence ⓘ

Sequence type

☒ Protein ☐ DNA ☐ RNA

Paste your sequence here - or use the example sequence

```
>sp|Q74700|TIM9_YEAST Mitochondrial import inner membrane translocase subunit TIM9 OS=Saccharomyces cerevisiae [strain ATCC 204508 / S288c]
OX=559292 GN=TIM9 PE=1 SV=1
MDALNSKEQQEFQKVVEQKQMKDFMRLYSNLVERCFTDCVNDFTTSKLTNKEQTCIMKCS
EKFLKHSERVGQRFOEQNAALGQGLGR
>sp|Q9Y5J7|TIM9_HUMAN Mitochondrial import inner membrane translocase subunit Tim9 OS=Homo sapiens OX=9606 GN=TIMM9 PE=1 SV=1
MAAQIPESDIKQKFEELGTYNKLTETCFLDCVKDFTTREVKPEETTCSEHCLOKYLKMT
QRISMRFQEHIOQNEALAAKAGLLGQPR
```

Choose file No file chosen

Use the example Clear sequence More example inputs

Tool output

CLUSTAL O(1.2.4) multiple sequence alignment

Download

```
sp|Q9XGX9|TIM9_ARATH MDASMMAGLDGLPEEDKAKMASMIDQLRDSLRMYNSLVERCFVDCVDSFTRKSLQKQE 60
sp|Q9XGX7|TIM9_ORYSJ MDKSMGLDGLPEEDKMRMAAMVDQLIRDSLRMYNSLVERCFDCVDTFRRKTLDKQE 60
sp|Q8J1Z1|TIM9_NEUCR -----MDGLTAAESRELDQRLQKRQVKEFMSVFGNLVDCFTACVDDFTSKALSGRE 52
sp|Q4WIQ2|TIM9_AS PFU -----MDGLTAAEQRELASRMERKQLKEFMTMYSKLVQRCDNCVNDFTTKSLISRE 52
sp|Q59R24|TIM9_CANAL -----MDQLNVKEQQEFQQLVEQKQMKDFMRLYSNLVSRCFDDCVNDFTSNLTSKE 52
sp|Q6BU42|TIM9_DEBHA -----MDQLNVKEQQDFQQLVEQKQMKDFMRLYSNLVSKCFDDCVNDFTSNLTTKE 52
sp|Q6C6Z2|TIM9_YARLI -----MDQLNNREQQEFQQLVEQKQMKDFMRLYSNLVSRCFDCVNDFTSKALSSRE 52
sp|Q74700|TIM9_YEAST -----MDALNSKEQQEFQQLVEQKQMKDFMRLYSNLVSRCFDCVNDFTSKLTNKE 52
sp|Q6FRT3|TIM9_CANGA -----MDQLNAKEQQEFQQLVEQKQMKDFMRLYSGLVERCFDCVNDFTSKLTSKE 52
sp|Q6CM57|TIM9_KLULA -----MDQLNGKEQQEFQQLVEQKQMKDFMRLYSNLVERCFSDCVNDFTSAKLTKE 52
sp|Q75750|TIM9_EREGS -----MDALNSREQQEFQQLVEQKQMKDFMRLYSNLVERCFSDCVNDFTSSKLTKE 52
sp|Q17754|TIM9_CAEEL -----MTSEQNIQTFRDFTQYNLVAEQCFNSCVNEFGSRTVSGKE 41
sp|Q9VYD7|TIM9_DROME -----MAKTPENIAIDQLDKDQIKTFSDFLMSYNKLTSETCTDCIRDFTTRDVKDSE 52
sp|Q9W762|TIM9_DANRE -----MAAQVTESDQIKQKFEFLGTYNKLTENCFMDCVKDFTTREVKPEE 45
sp|Q5ZIR8|TIM9_CHICK -----MAGQISETDQIKQKFEFLGTYNKLTENCFCMDCIRDFTTREVKPEE 45
sp|Q4V7R1|TIM9_XENLA -----MAAQMSSEDQIKQKFEFLGTYNKLTENCFLDCVKDFTTREVKAEE 45
sp|Q9WV98|TIM9_MOUSE -----MAAQIPESDQIKQKFEFLGTYNKLTETCFLDCVKDFTTREVKPEE 45
sp|Q9Y5J7|TIM9_HUMAN -----MAAQIPESDQIKQKFEFLGTYNKLTETCFLDCVKDFTTREVKPEE 45
sp|Q2KIV2|TIM9_BOVIN -----MAAQIPESDQIKQKFEFLGTYNKLTETCFLDCVKDFTTREVKPEE 45
sp|Q9WV97|TIM9_RAT -----MAAQIPESDQIKQKFEFLGTYNKLTETCFLDCVKDFTTREVKPEE 45
: : : : : ** * : *
```

```
sp|Q9XGX9|TIM9_ARATH ETCVMRCAEFLKHTMRVGMFAELNQNAPTQD----- 93
sp|Q9XGX7|TIM9_ORYSJ ESCVRRCAEFLKHSRMVGMFAELNQG VATPD----- 93
sp|Q8J1Z1|TIM9_NEUCR SGCISRCVLKSMSTQTRLGERFGE LNAAMTAE MQ-----RR----- 88
sp|Q4WIQ2|TIM9_AS PFU EGCIMRCVDKYM KASSRLNERFQEQAAMMQGGQ-----LPGR----- 90
sp|Q59R24|TIM9_CANAL TSCIAKCSEKFLKHSERVGQRFQE QNALLMQGGP-----K----- 87
sp|Q6BU42|TIM9_DEBHA TGCITKCSEKFLKHSERVGQRFQE QNALLMQNMQ-----KR----- 88
sp|Q6C6Z2|TIM9_YARLI ESCLEKCESEKFLKHSERVGQRFQE QNALLMQKR----- 85
sp|Q74700|TIM9_YEAST QTCIMKCSEKFLKHSERVGQRFQE QNAALGQGLG-----R----- 87
sp|Q6FRT3|TIM9_CANGA ESCILKCSEKFLKHSERVGQRFQE QNAALGQGLG-----R----- 87
sp|Q6CM57|TIM9_KLULA QNCIMRCSEKFLKHSERVGQRFQE QNAAMNQTMG-----R----- 87
sp|Q75750|TIM9_EREGS QTCIMRCSEKFLKHSERVGQRFQE QNAALNQSMG-----R----- 87
sp|Q17754|TIM9_CAEEL ESCANNCLDKFLKMTQRVSQR FQEHQLLNAQANGAAIKVENG GKINKIQ 90
sp|Q9VYD7|TIM9_DROME EKCSLNCMEKYLKMNQRVSQR FQEFQVIAHENALAM--AQKTGKL--- 95
sp|Q9W762|TIM9_DANRE TTCSESLCKYLKMTQRISMRFQEYHIQQNERWP-----QKPDY--- 84
sp|Q5ZIR8|TIM9_CHICK ITCSEHCLKYLKMTQRISMRFQEYHIQQNEALA-----AKAGLLSQPR 89
sp|Q4V7R1|TIM9_XENLA MTCSEHCLKYLKMTQRISMRFQEYHIQQNEALA-----AKAGLLGQPR 89
sp|Q9WV98|TIM9_MOUSE VTCSEHCLKYLKMTQRISMRFQEYHIQQNEALA-----AKAGLLGQPR 89
sp|Q9Y5J7|TIM9_HUMAN TTCSEHCLKYLKMTQRISMRFQEYHIQQNEALA-----AKAGLLGQPR 89
sp|Q2KIV2|TIM9_BOVIN TTCSEHCLKYLKMTQRISMRFQEYHIQQNEALA-----AKAGLLGQPR 89
sp|Q9WV97|TIM9_RAT VTCSEHCLKYLKMTQRISMRFQEYHIQQNEALA-----AKAGLLGQPR 89
* * * : . * : * * :
```

we can see
mammals
have similar
alignment

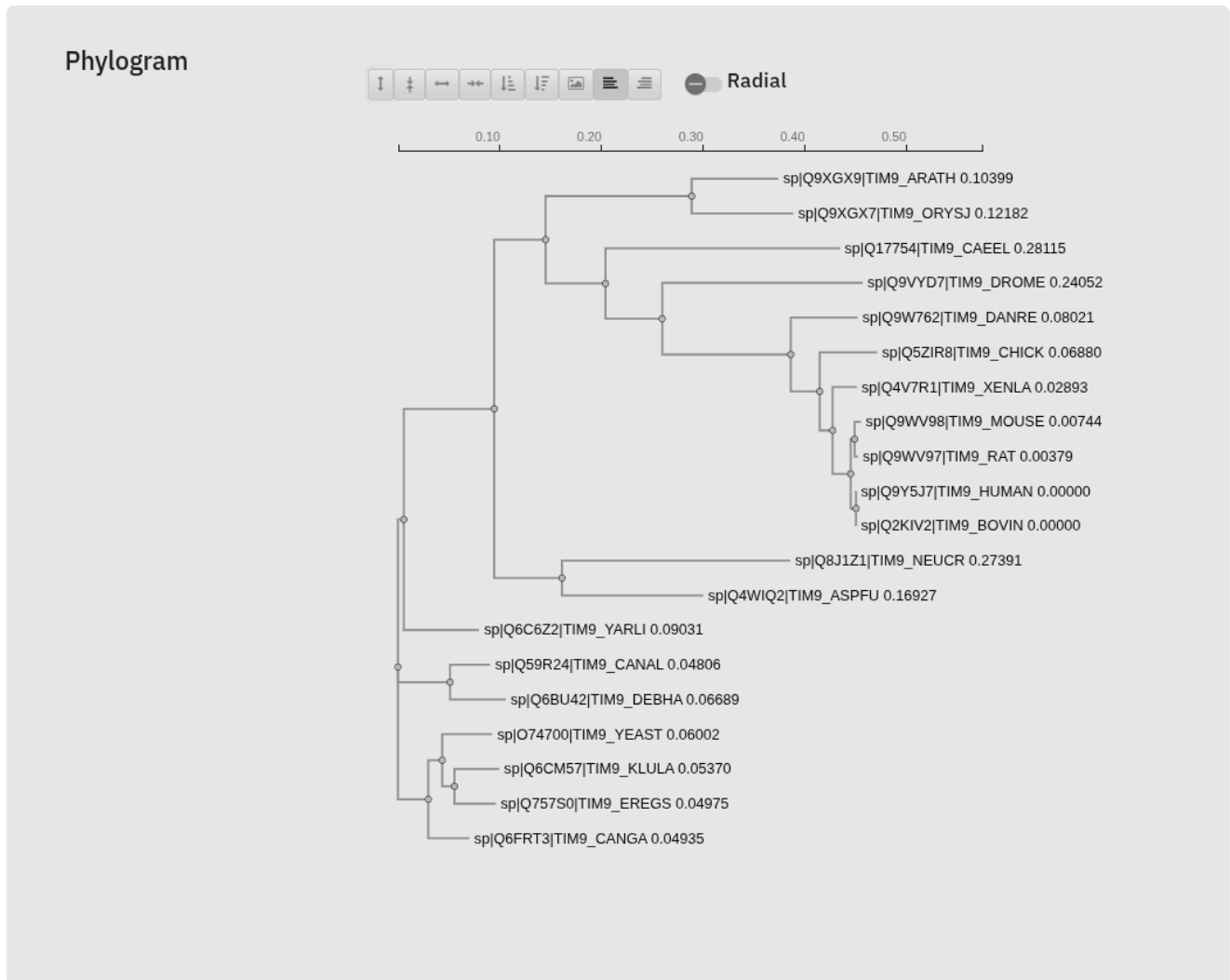
20 sequences

SP|Q9XGX9|TIM9_ARATH
SP|Q9XGX7|TIM9_ORYSJ
SP|Q8J1Z1|TIM9_NEUCR
SP|Q4WIQ2|TIM9_AS PFU
SP|Q59R24|TIM9_CANAL
SP|Q6BU42|TIM9_DEBHA
SP|Q6C6Z2|TIM9_YARLI
SP|Q74700|TIM9_YEAST
SP|Q6FRT3|TIM9_CANGA
SP|Q6CM57|TIM9_KLULA
SP|Q75750|TIM9_EREGS
SP|Q17754|TIM9_CAEEL
SP|Q9VYD7|TIM9_DROME
SP|Q9W762|TIM9_DANRE
SP|Q5ZIR8|TIM9_CHICK
SP|Q4V7R1|TIM9_XENLA
SP|Q9WV98|TIM9_MOUSE
SP|Q9Y5J7|TIM9_HUMAN
SP|Q2KIV2|TIM9_BOVIN
SP|Q9WV97|TIM9_RAT

1 10 20 30 40 50 60 70 80 90 100

```
MDASMMAGLDGLPEEDKAKMASMIDQLRDSLRMYNSLVERCFVDCVDSFTRKSLQKQEETCVMRCAEFLKHTMRVGMFAELNQNAPTQD-----
MDKSMGLDGLPEEDKMRMAAMVDQLIRDSLRMYNSLVERCFDCVDTFRRKTLDKQEESCVRRCAEFLKHSRMVGMFAELNQG VATPD-----
-----MDGLTAAESRELDQRLQKRQVKEFMSVFGNLVDCFTACVDDFTSKALSGRESGCISRCVLKSMSTQTRLGERFGE LNAAMTAE MQ-----
-----MDGLTAAEQRELASRMERKQLKEFMTMYSKLVQRCDNCVNDFTTKSLISREEGCIMRCVDKYM KASSRLNERFQEQAAMMQGGQ-----
-----MDQLNVKEQQDFQQLVEQKQMKDFMRLYSNLVSKCFDDCVNDFTSNLTTKETCITKCSEKFLKHSERVGQRFQE QNALLMQGGP-----
-----MDQLNNREQQEFQQLVEQKQMKDFMRLYSNLVSRCFDDCVNDFTSNLTSKETSCIAKCSEKFLKHSERVGQRFQE QNALLMQNMQ-----
-----MDALNSKEQQEFQQLVEQKQMKDFMRLYSNLVSRCFDCVNDFTSKLTNKEQTCIMKCSEKFLKHSERVGQRFQE QNAALGQGLG-----
-----MDQLNAKEQQEFQQLVEQKQMKDFMRLYSGLVERCFDCVNDFTSSKLTSKESCLEKCESEKFLKHSERVGQRFQE QNALLMQKR-----
-----MDQLNGKEQQEFQQLVEQKQMKDFMRLYSNLVERCFSDCVNDFTSAKLTKEQNCIMRCSEKFLKHSERVGQRFQE QNAAMNQTMG-----
-----MDALNSREQQEFQQLVEQKQMKDFMRLYSNLVERCFSDCVNDFTSSKLTSKETCITIMRCSEKFLKHSERVGQRFQE QNAALNQSMG-----
-----MTSEQNIQTFRDFTQYNLVAEQCFNSCVNEFGSRTVSGKEESCANNCLDKFLKMTQRVSQR FQEHQLLNAQANGAAIKVENG GKINKIQ
-----MAKTPENIAIDQLDKDQIKTFSDFLMSYNKLTSETCTDCIRDFTTRDVKDSEEKCSLNCMEKYLKMNQRVSQR FQEFQVIAHENALAM--AQKTGKL---
-----MAAQVTESDQIKQKFEFLGTYNKLTENCFMDCVKDFTTREVKPEETTCSESLCKYLKMTQRISMRFQEYHIQQNERWP-----QKPDY---
-----MAGQISETDQIKQKFEFLGTYNKLTENCFCMDCIRDFTTREVKPEEITCSEHCLKYLKMTQRISMRFQEYHIQQNEALA-----AKAGLLSQPR
-----MAAQMSSEDQIKQKFEFLGTYNKLTENCFLDCVKDFTTREVKAEEEMTCSEHCLKYLKMTQRISMRFQEYHIQQNEALA-----AKAGLLGQPR
-----MAAQIPESDQIKQKFEFLGTYNKLTETCFLDCVKDFTTREVKPEEVTTCSEHCLKYLKMTQRISMRFQEYHIQQNEALA-----AKAGLLGQPR
-----MAAQIPESDQIKQKFEFLGTYNKLTETCFLDCVKDFTTREVKPEETTCSEHCLKYLKMTQRISMRFQEYHIQQNEALA-----AKAGLLGQPR
-----MAAQIPESDQIKQKFEFLGTYNKLTETCFLDCVKDFTTREVKPEETTCSEHCLKYLKMTQRISMRFQEYHIQQNEALA-----AKAGLLGQPR
```

it even gives the phylogenetic tree



MAFFT

MAFFT (Multiple Alignment using Fast Fourier Transform) is a high speed multiple sequence alignment program. We have recently changed the default parameter settings for MAFFT. Alignments should run much more quickly and larger DNA alignments can be carried out by default. Please click the 'More options' button to review the defaults and change them if required. Important note: This tool can align up to 500 sequences or a maximum file size of 1 MB.

Input sequence ⓘ

Sequence type
☐ Automatic ☒ Protein ☐ DNA

Paste your sequence

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Please note that results can only be retrieved for jobs submitted within the last seven days.

Job ID: mafft-120260914-174427-0242-50206442-p1m

Choose file

Parameters

OUTPUT FORM
Pearson/FAST

More options ▼

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Title
MAFFT's job

Submit

RUNNING

Recent Jobs Close

Position (in Human Ref)	Ref Amino Acid	Clustal Omega Alignment	MAFFT Alignment	MUSCLE Alignment
Residue 1	Met1	Aligns with Met of chordates only; placed alongside empty gaps for plants, fungi, and worms.	Aligns directly with Met1 of yeast/fungi and Met1 of plants (MDAS...).	Shifted 7 residues downstream relative to plants; fails to align with plant/fungal Met1 .
Residue 5	Gln5	Aligns with internal Gln/Glu residues (Q24 of Arabidopsis, E16 of Yeast).	Flanked by open gap region; no alignment to plant or fungal sequences.	Aligns directly against T17 of Aspergillus and A17 of Neurospora.
Residue 60	His60	Aligns cleanly with H68 of Arabidopsis, H68 of Rice, and S59 of Neurospora.	Gapped in plants and fungi (pushed out into adjacent loop-gap column).	Aligns with H68 in plants, but gaps are distributed differently in <i>C. elegans</i> .
Residue 68	Gln68	Aligns with Gln across yeast/fungi (Q70) and Lys in <i>C. elegans</i> (K69).	Aligns with a 3-residue gap insertion (--) in all fungal and plant sequences.	Aligns against Ala70 in Arabidopsis and fungal Gln70 with no local gap shift.

Position (in Human Ref)	Ref Amino Acid	Clustal Omega Alignment	MAFFT Alignment	MUSCLE Alignment
Residue 72	Glu72	Aligns against Ala / Met in fungal sequences (A74 in Yeast).	Aligns precisely with Gln74 / Asn74 across yeast and plants (QNE motif).	Shifted upstream by 4 positions due to alternative gap penalties in chordates.

Q.8

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Exclude Optional
☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm
☒ blastp (protein-protein BLAST)
☐ PSI-BLAST (Position-Specific Iterated BLAST)
☐ PHI-BLAST (Pattern Hit Initiated BLAST)
☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)
Choose a BLAST algorithm [?](#)

BLAST
Search database **pdb** using **Blastp (protein-protein BLAST)**
☐ Show results in a new window

Note: Parameter values that differ from the default are highlighted in yellow and marked with ♦ s

Algorithm parameters

Feedback

since its a short sequence we need to change parameters slightly, default will give no similarity

Note: Parameter values that differ from the default are highlighted in yellow and marked with a sign

Algorithm parameters

Restore default search parameters

General Parameters

Max target sequences: 100
Select the maximum number of aligned sequences to display

Short queries: ☒ Automatically adjust parameters for short input sequences
Automatically adjust word size and other parameters to improve results for short queries.

Expect threshold: 1000

Word size: 2

Max matches in a query range: 0

Scoring Parameters

Matrix: PAM30

Gap Costs: Existence: 9 Extension: 1

Compositional adjustments: Conditional compositional score matrix adjustment

Filters and Masking

Filter: ☐ Low complexity regions

Mask: ☐ Mask for lookup table only
☐ Mask lower case letters

BLAST Search database **pdb** using **Blastp (protein-protein BLAST)**
☐ Show results in a new window

Feedback

- **Word size (2):** Lowers the seed threshold so short 14-residue queries can find initial matching.
- **PAM30 (instead of BLOSUM62):** more toward finding short, nearly exact sequence matches with higher relative information entropy per position.
- **High Expect Threshold (1000):** Prevents short alignments with naturally low bit scores from getting filtered out by random-chance statistical penalties.

Sequences producing significant alignments

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	Description ▾	Scientific Name ▾	Max Score ▾	Total Score ▾	Query Cover ▾	E value ▾	Per. Ident ▾	Acc. Len ▾	Accession
☒	Chain A, 1-deoxy-D-xylulose 5-phosphate reductoisomerase [...]	Escheric...	40.5	40.5	100%	8e-06	85.71%	406	1Q0H_A
☒	Chain A, 1-deoxy-D-xylulose 5-phosphate reductoisomerase [...]	Escheric...	40.5	40.5	100%	8e-06	85.71%	400	1JVS_A
☒	Chain A, 1-deoxy-D-xylulose 5-phosphate reductoisomerase [...]	Yersinia p...	39.2	39.2	86%	2e-05	91.67%	401	3IIE_A
☒	Chain A, 1-deoxy-D-xylulose 5-phosphate reductoisomerase [...]	Escheric...	34.6	34.6	100%	0.001	100.00%	398	1T1R_A
☒	Chain A, 1-deoxy-D-xylulose-5-phosphate reductoisomerase [...]	Escheric...	34.6	34.6	100%	0.001	100.00%	398	1K5H_A
☒	Chain A, 1-deoxy-D-xylulose 5-phosphate reductoisomerase [...]	Escheric...	34.6	34.6	100%	0.001	100.00%	424	2EGH_A
☒	Chain A, 1-deoxy-D-xylulose 5-phosphate reductoisomerase [...]	Escheric...	34.6	34.6	100%	0.001	100.00%	420	3ANL_A
☒	Chain A, 1-deoxy-D-xylulose 5-phosphate reductoisomerase [...]	Escheric...	34.6	34.6	100%	0.001	100.00%	406	1Q0L_A
☒	Chain A, 1-deoxy-D-xylulose 5-phosphate reductoisomerase [...]	Escheric...	34.6	34.6	100%	0.001	100.00%	410	3R0I_A
☒	Chain A, 1-deoxy-D-xylulose 5-phosphate reductoisomerase [...]	Acinetob...	34.1	34.1	93%	0.002	76.92%	406	4ZN6_A
☒	Chain A, 1-deoxy-D-xylulose 5-phosphate reductoisomerase [...]	Moraxella...	32.9	32.9	86%	0.004	75.00%	432	4ZQE_A
☒	Chain A, 1-deoxy-D-xylulose 5-phosphate reductoisomerase [...]	Moraxella...	32.9	32.9	86%	0.004	75.00%	415	4ZQG_A
☒	Chain A, 1-deoxy-D-xylulose 5-phosphate reductoisomerase [...]	Zymomo...	29.5	29.5	93%	0.078	84.62%	388	1R0K_A

Feedback

max score of 40.5 in E Coli so this means search reveals **no homologous counterpart** in the PDB; any local matches does not represent shared evolutionary ancestry or conserved 3D architecture.